

Problem Set 2: Loops and Conditional Logic

Prof. Jordan C. Hanson

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1 Exercise 1: Sorting with a For Loop

1. A placement test assigns each student a math score from 0 through 5. Students are assigned to courses according to

Score	Course
4, 5	Calculus
2, 3	Trigonometry
0, 1	Algebra

Begin with the following list of student tuples:

```
students = [  
    ("Aaliyah", 4),  
    ("Ben", 1),  
    ("Camila", 3),  
    ("Diego", 5),  
    ("Elena", 2),  
    ("Felix", 0),  
    ("Grace", 3),  
    ("Hector", 4),  
    ("Isabel", 1),  
    ("Jamal", 5),  
    ("Keira", 0),  
    ("Luis", 2)  
]
```

- Create the same three course structures used in the in-class exercise:

```
algebra = ["Algebra", []]
```

```
trigonometry = [  
    "Trigonometry", []  
]
```

```
calculus = ["Calculus", []]
```

Then combine them into the tuple

```
math_courses = (  
    algebra,  
    trigonometry,  
    calculus  
)
```

- Previously, the conditional statement had to be repeated once for every student. This time, use a **for** loop to examine each tuple in **students**:

```
for student in students:  
    name = student[0]  
    score = student[1]
```

```
# conditional logic goes here
```

Complete the loop using one **if-elif-else** statement.

- Use **append()** inside the conditional statement to place the student's name into the correct course. Do not manually type any student name into a course roster.

- After the loop finishes, print

```
print(math_courses)
```

and verify that all 12 students appear exactly once.

- Print the three rosters separately using indexing:

```
print(math_courses[0])  
print(math_courses[1])  
print(math_courses[2])
```

2 Exercise 2: Sorting with a While Loop

1. Repeat the placement problem from Exercise 1, but this time use a **while** loop instead of a **for** loop.

Begin with new empty course lists:

```
algebra2 = ["Algebra", []]
```

```
trigonometry2 = [  
    "Trigonometry", []  
]
```

```
calculus2 = ["Calculus", []]
```

and create

```
math_courses2 = (  
    algebra2,  
    trigonometry2,  
    calculus2  
)
```

- A **while** loop requires a variable that keeps track of which student is being processed. Begin with

```
i = 0
```

and construct a loop having the form

```
while i < 12:
    name = students[i][0]
    score = students[i][1]

    # conditional logic goes here

    i = i + 1
```

- Complete the same **if-elif-else** decision used in Exercise 1.

- Explain why

```
i = i + 1
```

must occur during every pass through the loop. What would happen if this statement were omitted?

- Print `math_courses2`. The student names in the three rosters should be identical to those obtained in Exercise 1.
- Compare the two approaches. In the **for** loop, what variable represents the current student? In the **while** loop, what variable determines which student is selected from the list?

3 Exercise 3: Math Placement System

1. Write a program called `math_placement.py` that collects student information from the user and automatically constructs the three course rosters.

The program will process exactly 10 students. You may assume that every score entered by the user is an integer from 0 through 5.

- Begin with an empty list:

```
students = []
```

Use a **while** loop and the `input()` function to collect the name and math score of each student. A useful starting structure is

```
i = 0
```

```
while i < 10:
    name = input("Student name: ")
    score = int(input("Math score: "))

    # store this student

    i = i + 1
```

- Store each student as a tuple of the form
`(name, score)`

and append that tuple to the `students` list.

After the loop finishes, the structure of `students` should be

```
[
    (name1, score1),
    (name2, score2),
    ...
    (name10, score10)
]
```

- Create the three initially empty course structures:

```
algebra = ["Algebra", []]
```

```
trigonometry = [
    "Trigonometry", []
]
```

```
calculus = ["Calculus", []]
```

and combine them into

```
math_courses = (
    algebra,
    trigonometry,
    calculus
)
```

- Use a **for** loop to process the completed `students` list. For every student, extract the name and score and use conditional logic to apply the placement rules

0, 1	Algebra
2, 3	Trigonometry
4, 5	Calculus.

The conditional logic, rather than manually written student names, must populate the three rosters.

- After all students have been sorted, print a final report having the general form

Math Placement Report

Algebra:

```
['student1', 'student2', ...]
```

Trigonometry:

```
['student3', 'student4', ...]
```

Calculus:

```
['student5', 'student6', ...]
```

Use the course names and student lists already stored inside `math_courses`; do not create a second copy of the roster information just for printing.

- Run the program with 10 test students. Your test data must include at least one score from each of the three placement ranges. Verify that

- every entered student appears in exactly one course,
 - scores 0–1 are assigned to Algebra,
 - scores 2–3 are assigned to Trigonometry, and
 - scores 4–5 are assigned to Calculus.
- Your completed program should use only concepts we have used in class: variables, strings and integers, `input()`, `print()`, lists, tuples, indexing, `append()`, conditional statements, and simple `for` and `while` loops.

Do not use dictionaries, functions, list comprehensions, or other Python structures that have not yet been introduced.